Supporting Information

Amplitude and midpoint statistics of the same chemotaxis cells demand incompatible receptor models

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October 2026

This Supporting Information documents the complete computational output of the accompanying repository (`chemotaxis-receptor-layer`, release v1.0.0): every analysis script, the numerical results each produced, and the provenance of the third-party FRET archive. Numbers are transcribed verbatim from `results/`; scripts are plain Python 3 (numpy/scipy/matplotlib) and read the `.mat` files in `data_external/moore2024_fret/` directly.

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1 Pipeline map: scripts and outputs

Script	Computes	Archived output
`code49_P7_blind_adjudication_FCD_Weber`	P7 blind adjudication of the fold-change/Weber claim on the Moore 2024 archive	adjudication tables
`code49b_P7_erratum_readjudication`	erratum re-adjudication (censoring-aware)	the code49b unit-table CSV (Chinese-named, `results/P7_Moore2024_FCD-Weber/`)
`code51chemo_receptor_layer_selfcheck`	receptor-layer internal-consistency self-check	console log
`code52_protocol_artifact_test_raw_FRET_coordinate_analysis`	protocol-artifact test in raw FRET coordinates	console/tables
`code53_CheA_node_mechanism_kernel_competition`	CheA-node mechanism: kernel competition	console/tables
`code54_K12_joint_fit_rupture_quantification`	joint-fit rupture quantification on the $K_{1/2}$ ladder	console/tables
`code55_multispecies_local_adaptation_joint_test`	multi-species / local-adaptation joint test	console/tables
`code56_FRET_readout_nonlinearity_test`	FRET readout-nonlinearity control	console/tables
`code57_TCS_depletion_robustness_gain_retuning`	TCS-depletion robustness with gain retuning	console/tables
`code58_P8_independent_arm_serine_Lasp`	P8 independent arm (serine, L-aspartate)	console/tables
`code59_residual_waterfall_window_structure_rhoF`	residual waterfall; window structure of ρ_F	console/tables
`code60_heterogeneous_population_censoring_test`	heterogeneous-population censoring control	console/tables
`code61_rowwise_Hill_steepness_test`	row-wise Hill-steepness test	console/tables
`code62_single_cell_same_source_statistics`	same-source single-cell statistics	console/tables
`code70_mixed_ligand_validation_arm_P1P2`	prospective mixed-ligand validation arms P1/P2	`chemotaxis_prospective_arm.csv`
`chemotaxis_adaptation_audit.py`	adaptation-relay baseline audit on the full archive	`chemotaxis_adaptation_audit.json`

2 P7 adjudication: $K_{1/2}$ medians vs the Weber line

Censoring-aware re-adjudication (code49b) of the FCD–Weber claim on the Moore et al. (2024) single-cell FRET archive. Table 2 is the archived unit table (verbatim rows).

The per-cell unit table (the unit-table CSV under `results/P7_Moore2024_FCD-Weber/` (Chinese-named)) carries, per (background B, foreground F) condition: fold change r, threshold T, $\log_{10} r$, $\log_{10} T$, n cells,

Table 2: $K_{1/2}$ medians per background level vs Weber-line prediction.

Background (μM)	$K_{1/2}$ median (μM)	IQR lo–hi	censored / n	Weber pred. (μM)	dev. (dex)
0.0	2.033	1.583–3.105	179/268	2.282	−0.050
0.01	2.896	2.080–3.065	69/101	2.293	+0.101
0.1	2.168	1.769–2.844	35/57	2.399	−0.044

and the two arm statistics R_a, R_A; e.g. B=0.01, F=0.2 μM: $r = 21.0$, $T = 0.21$, $n = 101$, $R_a = 0.0655$, $R_A = 0.0209$.

3 Adaptation-relay baseline audit

Full-archive audit (`chemotaxis_adaptation_audit.json`): 2217 cells, 77,595 response events; median baseline activity after high- vs low-amplitude preceding stimuli 0.434 vs 0.448; median absolute response 0.608. The per-(preceding-dose) baseline bands are archived in the same JSON.

4 Prospective mixed-ligand arms (code70)

Frozen-then-measured prospective arm V1 (clause P-1b, amplitude table of the L-aspartate component). Verbatim rows of `chemotaxis_prospective_arm.csv`:

Table 3: Prospective arm V1: measured vs predicted amplitudes (tolerance: max |dev| = 0.08).

Foreground (μM)	measured	predicted	deviation	hit
5	0.047	0.119	−0.072	Y
10	0.085	0.152	−0.067	Y
20	0.139	0.187	−0.048	Y

5 Third-party data provenance

`data_external/moore2024_fret/` mirrors, verbatim, the 48-file single-cell FRET archive of Moore, Kamino, Kottou, Shimizu & Emonet, *Cell Syst.* 15, 628–638 (2024), Dryad doi:10.5061/dryad.nvx0k6dzz, under its CC0 1.0 public-domain dedication; please cite the original study. Receptor-layer parameter priors derive from the Kamino/Shimizu chemoreceptor literature (incl. *Proc. Natl Acad. Sci. USA* 108, 13870 (2011)); fold-change readout evidence from Lazova et al., *Proc. Natl Acad. Sci. USA* 108, 13870–13875 (2011). Full notes in `data_external/PROVENANCE.md`.